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EffiNIDS: A Computation-Aware Feature Selection Approach for High-Performance Network Intrusion Detection Systems

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Extended Abstract

Research Area

Network Intrusion Detection Systems (NIDS) analyze network traffic to identify malicious activities and potential security threats. However, processing a large number of traffic features can increase computational requirements, particularly during model training and prediction. This creates challenges for deploying intrusion detection in resource-constrained environments. This study introduces ***EffiNIDS***, a feature-selection-based evaluation framework that investigates whether reducing the number of input features can lower the computational cost of Random Forest-based network intrusion detection while maintaining comparable detection performance. The framework is experimentally evaluated using the UNSW-NB15 dataset.

Goals/Objectives

The primary objective of ***EffiNIDS*** is to evaluate the trade-off between feature reduction, detection performance, and computational cost. It compares a full-feature baseline with two feature-selection approaches: Mutual Information (MI) and Random Forest (RF) impurity-based feature importance. Each selection method retains 97 of the 194 encoded features, representing a 50% reduction. The evaluation examines whether this reduction can maintain detection performance while decreasing inference time, and whether the additional cost of feature selection affects the overall training process. This is important for developing computationally practical intrusion detection approaches for environments with limited resources.

Methodology

Experiments were conducted using the official UNSW-NB15 training and testing files, containing 175,341 and 82,332 records, respectively, with 45 columns each. The identifier (id) was excluded, and the multiclass attack category (attack_cat) was removed to avoid potential target leakage. This left 42 input predictors, comprising 39 numerical and three categorical features, alongside the binary target label. One-hot encoding of the categorical features produced 194 model input features.

Three configurations were evaluated using the same Random Forest classifier with 200 trees, max_features="sqrt", and random_state=42. The baseline used all 194 encoded features. The MI configuration selected the top 97 features according to their mutual information with the target, while the RF-importance configuration selected the top 97 using impurity-based feature importance. MI measures the statistical dependence between individual features and the target, whereas RF impurity importance estimates feature contributions to impurity reduction across the trees.

All preprocessing and feature-selection operations were fitted exclusively on training data. Each configuration was evaluated on the same official test set. Detection performance was measured using accuracy, attack F1-score, macro-F1, and false-positive rate (FPR). Computational cost was assessed through total training and selection time and full-test inference time. Training and selection time included all operations required to produce the final model and was measured once per configuration. Inference time included preprocessing and prediction over the complete test set, with the median of three runs reported. Experiments were conducted on a Windows 11 system equipped with an AMD Ryzen 5 5600G processor and 16 GB RAM.

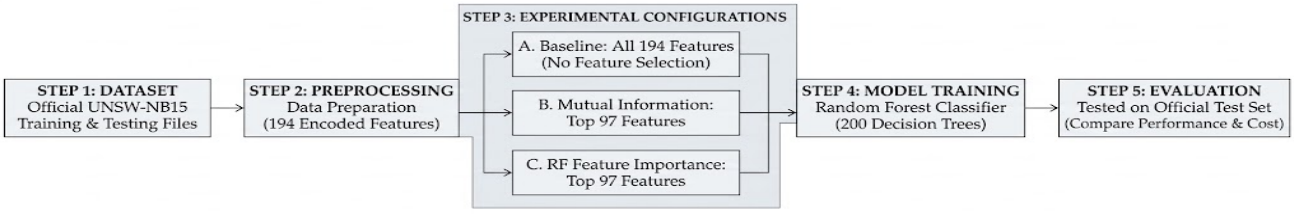


Fig. 1. Experimental pipeline. Preprocessing and selection were fitted on training data only; the baseline skips the selection step.

Results and Analysis

Table I: On the official UNSW-NB15 test set, the observed performance estimates were numerically similar (accuracy: 87.09–87.19%; attack F1: 89.38–89.45%; macro-F1: 86.46–86.58%). MI and RF-importance selection reduced full-test inference time by 15.1% and 8.1%, respectively, while increasing reported training-plus-selection time by 22.8% and 103.6%. Because model-building time was measured once and only one model seed was evaluated, these results are descriptive; they do not establish statistical equivalence or generalizable runtime savings. The FPR remained approximately 27%.

TABLE I. COMPARATIVE PERFORMANCE AND COMPUTATIONAL COST OF THE EVALUATED CONFIGURATIONS

Configuration	Features	Accuracy (%)	Attack F1 (%)	Macro-F1 (%)	FPR (%)	Training + Selection (s)	Inference (s)
Baseline (All Features)	194	87.09	89.38	86.46	27.05	61.3	0.516
Mutual Information (MI)	97	87.15	89.41	86.53	26.82	75.3	0.438
RF Feature Importance	97	**87.19**	**89.45**	**86.58**	**26.73**	124.8	0.474

A supplementary post-hoc predictor-pattern analysis examined test records whose exact predictor combinations had appeared in the training data. Among 82,332 test records, 8,015 had previously observed combinations associated with consistent labels, and all three models classified these records correctly. Another 526 records had combinations associated with conflicting labels across the combined training and test data. Accuracy on this small subset ranged from 9.1% to 9.5%. This diagnostic suggests that contradictory labels for identical observed predictors may present a challenge for classification. However, it is exploratory and does not establish that the underlying records are incorrectly labelled.

The main strength of ***EffiNIDS*** is its comparative evaluation of detection performance alongside the computational overhead of feature selection, rather than considering feature reduction alone. The findings indicate that retaining half of the encoded features maintained comparable detection performance and reduced inference time, but increased total training and selection time. These measurements are hardware-dependent, and training and selection times were based on single runs. Moreover, runtime measurements do not directly establish energy savings.

Future work will investigate multiple random seeds and feature subset sizes, include random feature selection as a control, separate feature-selection time from final model training time, and evaluate additional classifiers and datasets. Repeated timing measurements and memory-usage analysis will also help establish the reliability and practical applicability of the framework.

References

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Keywords: EffiNIDS, network intrusion detection, feature selection, Random Forest, computational efficiency.